Total No. of Questions : 6]

SEAT No. :

P5446

[Total No. of Pages : 3

B.E./Insem/Oct.-16
B.E. (Mechanical)
DYNAMICS OF MACHINERY
(2012 Pattern) (Semester - I)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates :

1) *Solve Q1 or Q2, Q3 or Q4, Q5 or Q6.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume suitable data, if necessary.*

Q1) a) Explain the concept of single plane and multiple plane balancing. **[4]**

b) A four cylinder vertical engine has cranks 150 mm long. The cylinders are spaced 200 mm apart. Mass of reciprocating parts of 1st, 2nd and 4th cylinders are 50 kg, 60 kg and 50 kg respectively. Find the reciprocating mass of the 3rd cylinder and relative angular positions of the cranks to achieve complete primary balance. **[6]**

OR

Q2) a) Explain the terms static balancing and dynamic balancing. **[4]**

b) A line shaft carries four pulleys P, Q, R and S, spaced equally at 0.6 m apart and at radii 0.1 m, 0.225 m, 0.15 m and 0.15 m respectively. The mass of the pulleys Q, R and S is 10 kg, 5.5 kg and 3.6 kg respectively. If the shaft is to be in complete balance, determine : **[6]**

i) Mass of Pulley P

ii) Angular positions of all four pulleys

Q3) a) The motion of a point is given by $\alpha = -9x$ where α and x and the acceleration and displacement of SHM. The amplitude is 0.05m. Find **[4]**

i) The time period

ii) Frequency

iii) Displacement and velocity after 20 seconds.

P.T.O.

b) Determine the natural frequency of the system shown in Fig. Q.3. B. **[6]**

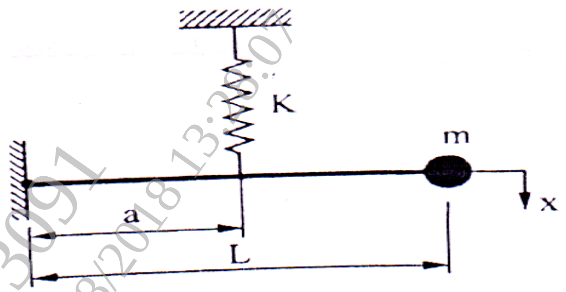

Fig.Q.3.B

OR

Q4) a) With neat sketch explain under damped system, over damped system and critically damped system? Give a practical application of each. if so. **[6]**

b) A mass of 3 kg is supported on an isolator having a spring constant of 3000 N/m and viscous damping. If the amplitude of free vibration of the mass falls to one half its original value in 2 sec, determine the damping coefficient of the isolator. **[4]**

Q5) a) Fill in the blanks : **[4]**

i) Rotating shafts tend to vibrate violently in transverse direction at certain speed. This speed is called ________.

I) critical speed

II) whirling speed

III) whipping speed

IV) all of the above

ii) The angle between spring force and damping force is _____

I) 90°

II) 180°

III) 0°

IV) none of the above

iii) For shaft speed less than the critical speed, the phase difference between displacement and centrifugal force is ______.

I) 0°

II) 45°

III) 90°

IV) 180°

iv) The factor which affects the critical speed of a shaft is ______.

I) eccentricity

II) diameter of the disc

III) span of the shaft

IV) all above

b) A single cylinder vertical petrol engine of total mass 320 kg is mounted on a steel chassis and causes a vertical static deflection of 2 mm. The reciprocating parts of the engine have a mass of 24 kg and move through a vertical stroke of 150 mm with SHM. A dashpot attached to the system offers a resistance of 490 N at a velocity of 0.3 m/s. Determine : **[6]**

i) the speed of driving shaft at resonance

ii) the amplitude of steady state vibrations when the driving shaft of the engine rotates at 480 rpm.

OR

Q6) a) Plot frequency response curves and draw any four conclusions from it. **[4]**

b) A tractor driving across a field that has undulations at regular intervals. The distance between the bumps is about 4.2 m. The driver's seat is attached to a spring to absorb some of shock as the tractor moves over rough ground. Assume the spring constant to be 2.0×10^4 N.m^{-1} and the mass of the seat to be 50 kg and the mass of the driver 70 kg. The tractor is driven at 30 km.h^{-1} over the undulations. Will an accident occur? **[6]**